

# Figure 3

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#ENV

```
rm(list = ls())
setwd("/Users/liqun/Desktop/scMutrace-tutorial/ReproducibilityAnalysis/Figure3/")
```

## 1 Libraries

```
library(ggplot2)
library(ggpubr)
library(ggrepel)
library(dplyr)
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##      filter, lag
## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union
library(tidyr)
library(ComplexHeatmap)
## Loading required package: grid
```

```
## =====
## ComplexHeatmap version 2.24.1
## Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
## Github page: https://github.com/jokergoo/ComplexHeatmap
## Documentation: http://jokergoo.github.io/ComplexHeatmap-reference
##
## If you use it in published research, please cite either one:
## - Gu, Z. Complex Heatmap Visualization. iMeta 2022.
## - Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
##   genomic data. Bioinformatics 2016.
##
##
## The new InteractiveComplexHeatmap package can directly export static
## complex heatmaps into an interactive Shiny app with zero effort. Have a try!
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(ComplexHeatmap))
## =====
library(circlize)
## =====
## circlize version 0.4.16
## CRAN page: https://cran.r-project.org/package=circlize
## Github page: https://github.com/jokergoo/circlize
## Documentation: https://jokergoo.github.io/circlize\_book/book/
##
## If you use it in published research, please cite:
## Gu, Z. circlize implements and enhances circular visualization
##   in R. Bioinformatics 2014.
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(circlize))
## =====
```

## 2 Load Data

```
# XVSeq
# Ultraviolet radiation shapes dendritic cell leukaemia transformation in the skin
# https://www.nature.com/articles/s41586-023-06156-8
# Mutation_sample_XVSeq <- read.csv("../Data/XVSeq/Mutation_sample.csv", row.names = 1)
# Mutation_sample_XVSeq[is.na(Mutation_sample_XVSeq)] <- 0
# Mutation_sample_XVSeq$Gene <- rownames(Mutation_sample_XVSeq)

# GoTSeq
# Deconvoluting clonal and cellular architecture in IDH-mutant acute myeloid leukemia
# https://www.cell.com/cell-stem-cell/fulltext/S1934-5909\(25\)00179-1
# Mutation_sample_GoTSeq <- read.csv("../Data/GoT/Mutation_sample.csv", row.names = 1)
# Mutation_sample_GoTSeq[is.na(Mutation_sample_GoTSeq)] <- 0
# Mutation_sample_GoTSeq$Gene <- rownames(Mutation_sample_GoTSeq)

# SMARTSeq
# Therapy-Induced Evolution of Human Lung Cancer Revealed by Single-Cell RNA Sequencing
```

```

# https://www.sciencedirect.com/science/article/pii/S0092867420308825
# Mutation_sample_SMARTSeq <- read.csv("../Data/SMARTSeq/Mutation_sample.csv", row.names = 1)
# Mutation_sample_SMARTSeq[is.na(Mutation_sample_SMARTSeq)] <- 0
# Mutation_sample_SMARTSeq$Gene <- rownames(Mutation_sample_SMARTSeq)

# Merge
# Mutation_sample_XVSeq <- Mutation_sample_XVSeq |> dplyr::rename_with(~ paste0("XVSeq_", .), -Gene)
# Mutation_sample_GoTSeq <- Mutation_sample_GoTSeq |> dplyr::rename_with(~ paste0("GoTSeq_", .), -Gene)
# Mutation_sample_SMARTSeq <- Mutation_sample_SMARTSeq |> dplyr::rename_with(~ paste0("SMARTSeq_", .),

# Merged_mutation <- Mutation_sample_XVSeq %>%
#   dplyr::full_join(Mutation_sample_GoTSeq, by = "Gene") %>%
#   dplyr::full_join(Mutation_sample_SMARTSeq, by = "Gene")

# Merged_mutation[is.na(Merged_mutation)] <- 0

# save merge data
# write.csv(Merged_mutation, "../Data/0_Mutation_sample_merge.csv")

# load Mutation sample matrix
Merged_mutation <- read.csv("../Data/MergeData/0_Mutation_sample_merge.csv")

# load Mutant cell fraction
Mutation_fraction <- read.csv("../Data/MergeData/0_Benchmarking_MutCellFrac.csv", header = T)

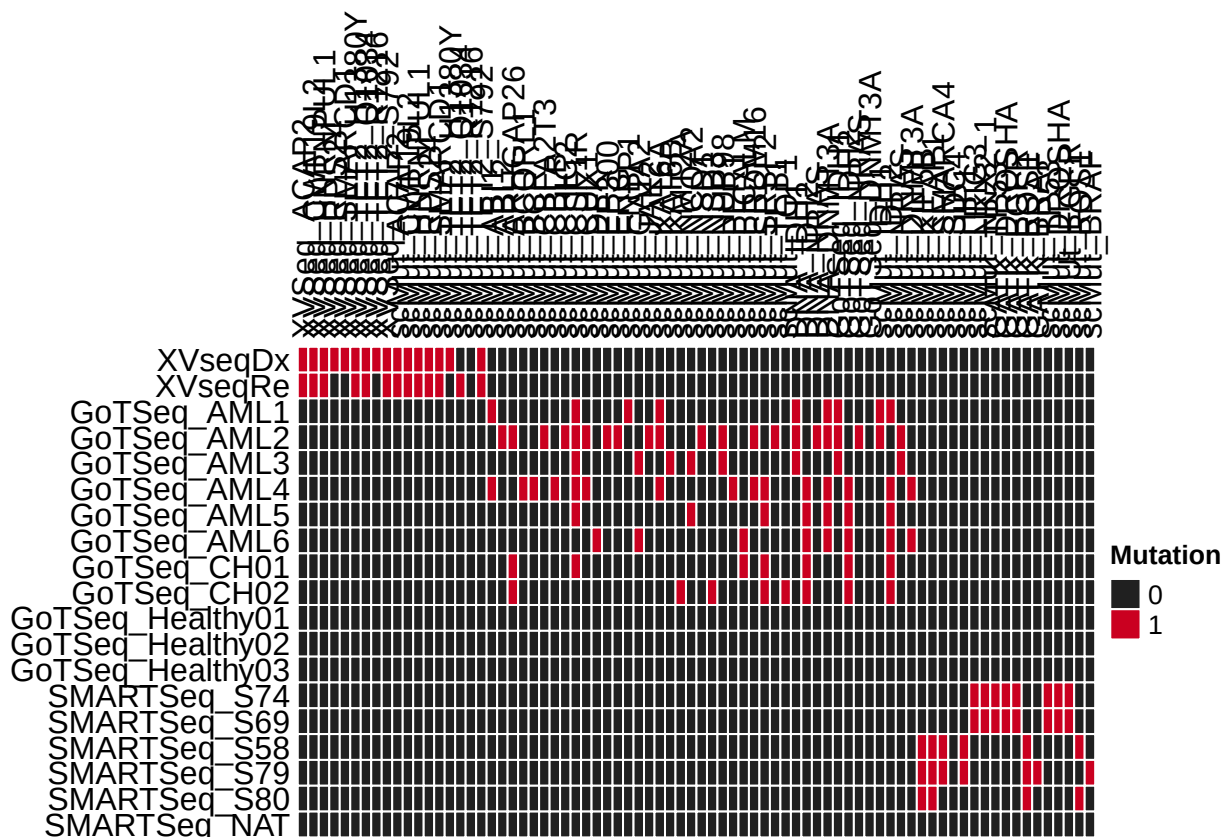
rownames(Merged_mutation) <- Merged_mutation$Gene
Merged_mutation_Ann <- Merged_mutation[, c("Gene", "Tag")]
Merged_mutation$Gene <- NULL
Merged_mutation$Tag <- NULL

# heatmap
mat <- as.matrix(Merged_mutation)
bin_mat <- ifelse(mat > 0, 1, 0)

col_fun2 <- c("0" = "#202020",
              "1" = "#ca0020")

Heatmap(
  t(bin_mat),
  name = "Mutation",
  col = col_fun2,
  cluster_rows = FALSE,
  cluster_columns = FALSE,
  row_names_side = "left",
  column_names_side = "top",
  rect_gp = gpar(col = "white", lwd = 1)
)

```



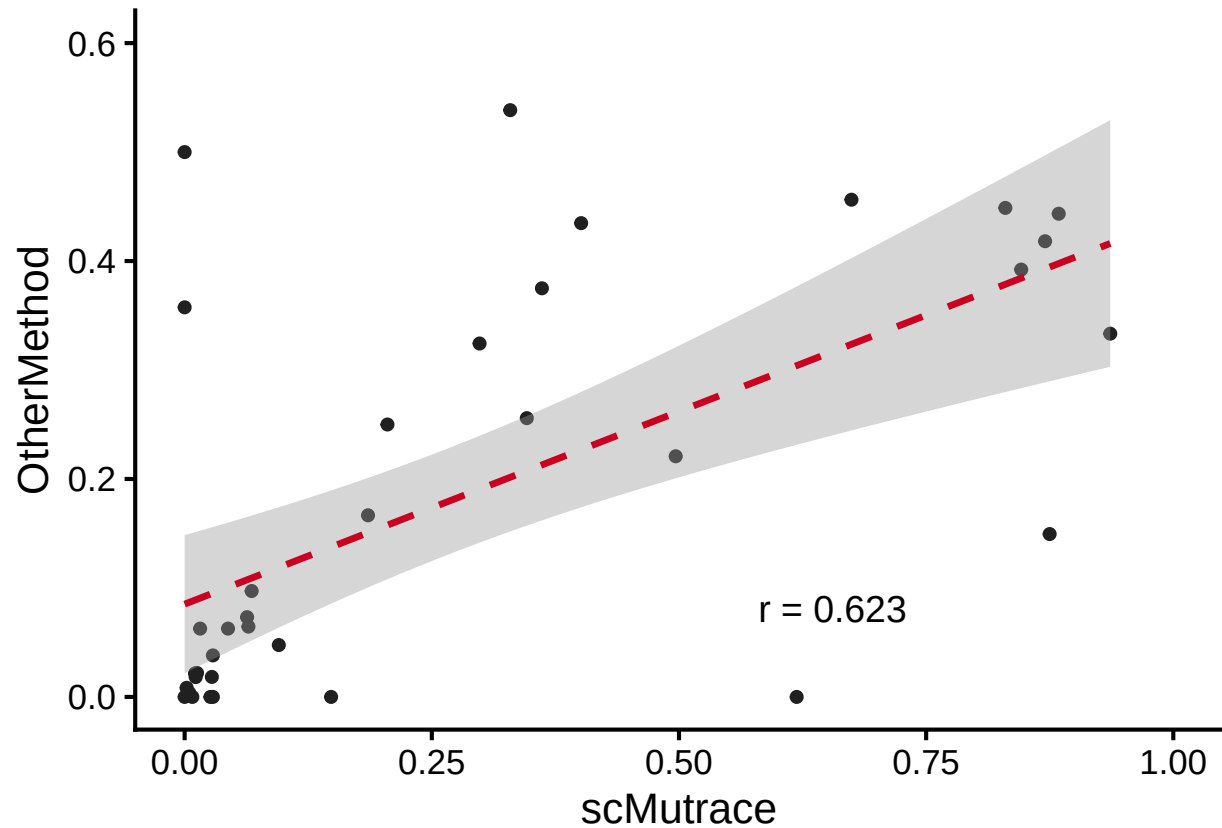
```
p_correlation_scMut_OtherMethods <- ggplot(Mutation_fraction, aes(x = scMutrace, y = OtherMethod)) +
  geom_point(size = 1.5, color = "#202020") +
  geom_smooth(method = "lm", se = TRUE, color = "#ca0020", linetype = "dashed", lwd = 1.2) +
  theme_classic(base_size = 16) +
  xlab("scMutrace") +
  ylab("OtherMethod") +
  xlim(0, 1) + ylim(0, 0.6) +
  annotate("text",
    x = max(Mutation_fraction$scMutrace) * 0.7,
    y = max(Mutation_fraction$OtherMethod) * 0.15,
    label = paste0("r = ", round(cor(Mutation_fraction$scMutrace, Mutation_fraction$OtherMethod),
    size = 5))

cor.test(Mutation_fraction$scMutrace, Mutation_fraction$OtherMethod, method = "pearson")
##
## Pearson's product-moment correlation
##
## data: Mutation_fraction$scMutrace and Mutation_fraction$OtherMethod
## t = 4.7132, df = 35, p-value = 3.803e-05
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3747529 0.7880265
## sample estimates:
## cor
```

```
## 0.6231096
```

```
p_correlation_scMut_OtherMethods
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
# clean mat  
rm(mat, bin_mat)
```

### 3 benchmarking

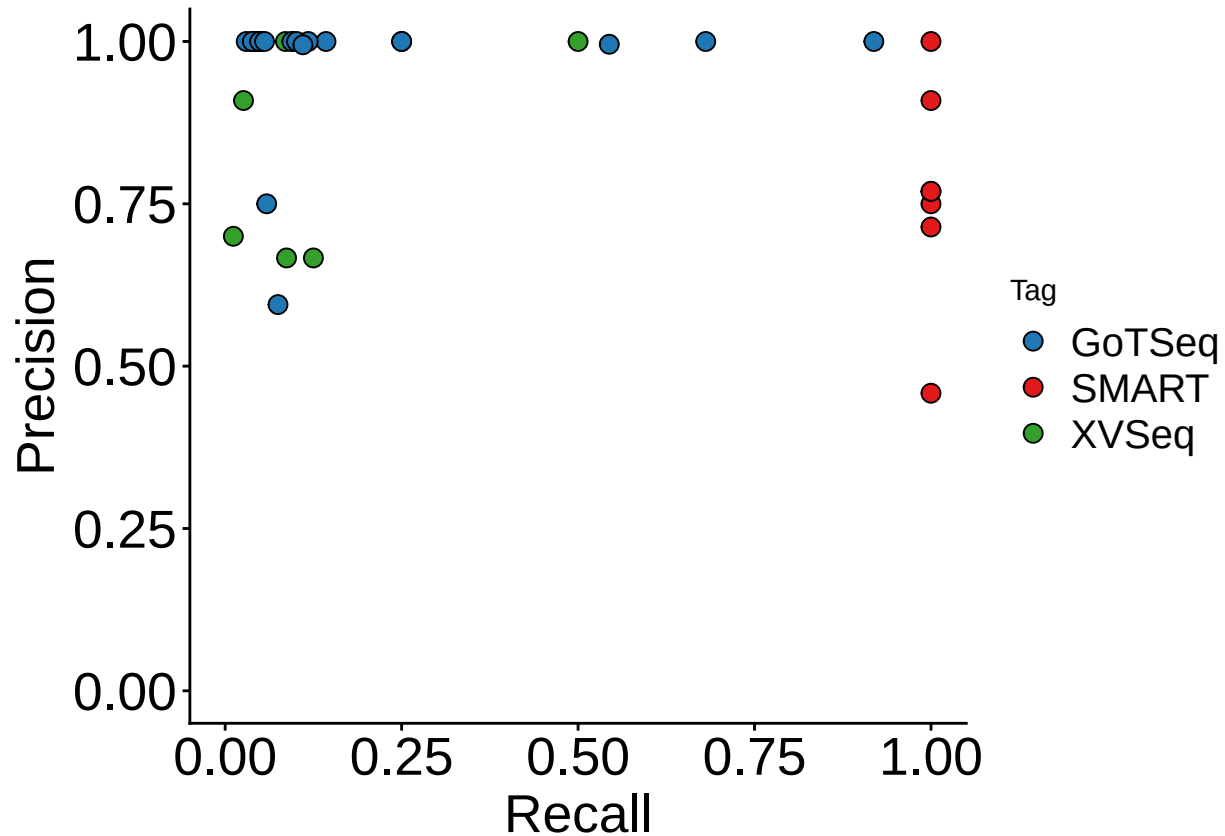
```
benchmarking_data <- read.csv("../Data/MergeData/0_Benchmarking_Data.csv", header = T)
```

```
p_benchmarking_res <- ggplot(data = benchmarking_data) +  
  geom_point(mapping = aes(x = Recall, y = Precision, fill = Tag),  
    color = "black", shape = 21, size = 3, show.legend = TRUE) +  
  scale_fill_manual(values = c("#1f78b4", "#e31a1c", "#33a02c")) +  
  xlab("Recall") + ylab("Precision") + xlim(0, 1) + ylim(0, 1) +  
  theme_classic() +  
  theme(  
    axis.title = element_text(size = 20),  
    axis.text.y = element_text(size = 20),  
    axis.text.x = element_text(size = 20),
```

```

    legend.text = element_text(size = 15)
  )
p_benchmarking_res

```



## 4 GO

```

GO_mutation <- read.csv("../Data/MergeData/1_GO.csv", header = T)

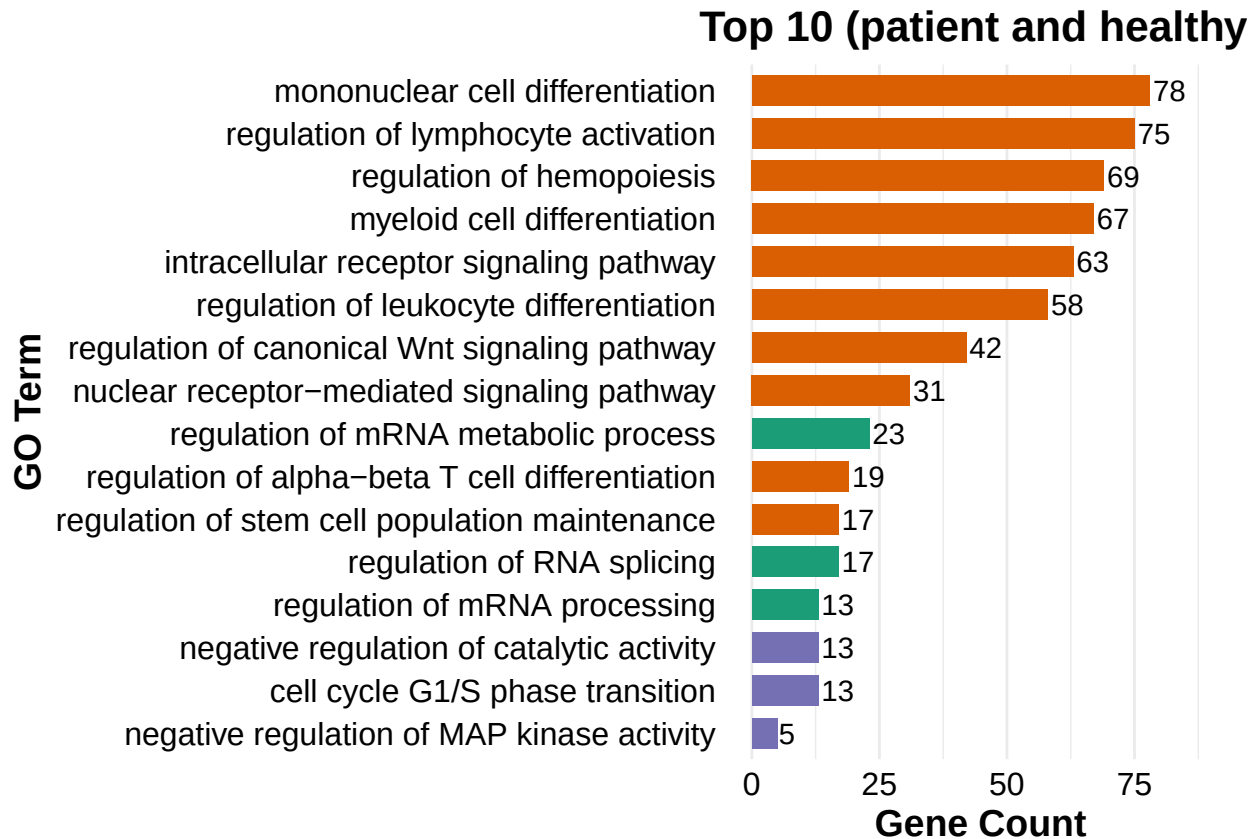
p_GOs_mutation <- ggplot(GO_mutation, aes(x = reorder(Description, Count), y = Count, fill = Type)) +
  geom_bar(stat = "identity", width = 0.7) +
  coord_flip() +
  geom_text(aes(label = Count), hjust = -0.1, size = 4, color = "black") +
  labs(
    x = "GO Term",
    y = "Gene Count",
    title = "Top 10 (patient and healthy)"
  ) + theme_minimal() +
  theme(
    axis.text = element_text(size = 12, color = "black"),
    axis.title = element_text(size = 14, face = "bold"),
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5),

```

```

legend.position = "none",
panel.grid.major.y = element_blank()
) + ylim(0, max(GO_mutation$Count)*1.15) +
scale_fill_manual(values = c(
  "AML" = "#d95f02",
  "CH" = "#1b9e77",
  "Healthy" = "#7570b3"))
p_GOs_mutation

```



## 5 SBS mutation signatures

```

SBS_data <- read.csv("../Data/MergeData/Signature_contributions_PDC.csv", header = T)
df_long <- SBS_data %>%
  tidyr::pivot_longer(cols = c(DX, RE),
    names_to = "Stage",
    values_to = "Contribution")

signature_colors <- c(
  "SBS1" = "#a6cee3",
  "SBS5" = "#1f78b4",
  "SBS7a" = "#e31a1c",

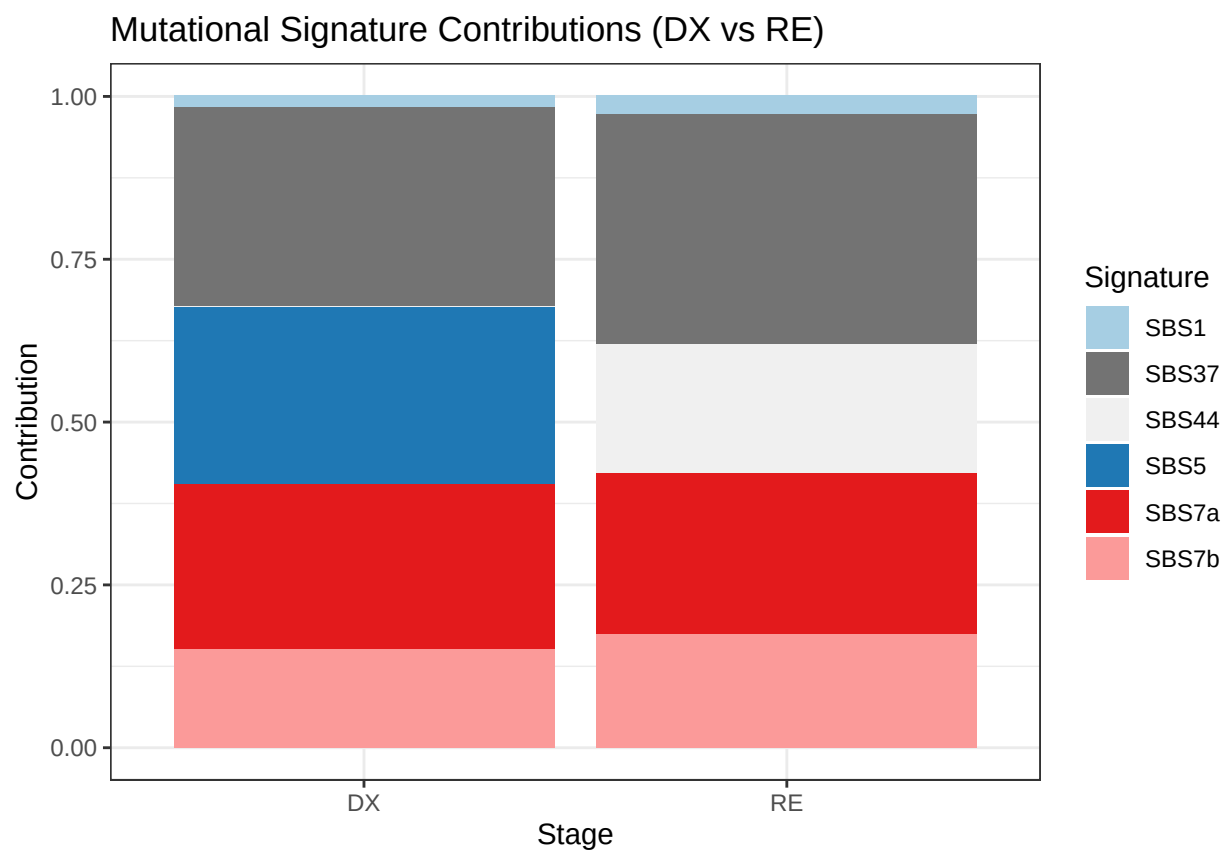
```

```

"SBS7b" = "#fb9a99",
"SBS37" = "#737373",
"SBS44" = "#f0f0f0"
)

p_SBS_XVSeq <- ggplot(df_long, aes(x = Stage, y = Contribution, fill = Signature)) +
  geom_bar(stat = "identity", position = "stack") +
  scale_fill_manual(values = signature_colors) +
  theme_bw() +
  labs(
    title = "Mutational Signature Contributions (DX vs RE)",
    x = "Stage",
    y = "Contribution"
  )
p_SBS_XVSeq

```



## 6 sessionInfo

```

sessionInfo()
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20

```



```

## Running under: macOS Tahoe 26.2
##
## Matrix products: default
## BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] grid      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] circlize_0.4.16      ComplexHeatmap_2.24.1  tidyr_1.3.1
## [4] dplyr_1.1.4          ggrepel_0.9.6          ggpubr_0.6.1
## [7] ggplot2_4.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6          shape_1.4.6.1          rjson_0.2.23
## [4] xfun_0.53             GlobalOptions_0.1.2    rstatix_0.7.2
## [7] lattice_0.22-7        Cairo_1.6-5            vctr_0.6.5
## [10] tools_4.5.1           generics_0.1.4         stats4_4.5.1
## [13] parallel_4.5.1        tibble_3.3.0           cluster_2.1.8.1
## [16] pkgconfig_2.0.3       Matrix_1.7-4           RColorBrewer_1.1-3
## [19] S7_0.2.0              S4Vectors_0.46.0       lifecycle_1.0.4
## [22] compiler_4.5.1        farver_2.1.2           codetools_0.2-20
## [25] carData_3.0-5         clue_0.3-66            htmltools_0.5.8.1
## [28] yaml_2.3.10           Formula_1.2-5          pillar_1.11.1
## [31] car_3.1-3             crayon_1.5.3           iterators_1.0.14
## [34] abind_1.4-8           foreach_1.5.2          nlme_3.1-168
## [37] tidyselect_1.2.1      digest_0.6.37          purrr_1.1.0
## [40] labeling_0.4.3        splines_4.5.1          fastmap_1.2.0
## [43] colorspace_2.1-2      cli_3.6.5              magrittr_2.0.4
## [46] dichromat_2.0-0.1     broom_1.0.10           withr_3.0.2
## [49] scales_1.4.0          backports_1.5.0        rmarkdown_2.30
## [52] matrixStats_1.5.0     ggsignif_0.6.4         png_0.1-8
## [55] GetoptLong_1.0.5      evaluate_1.0.5         knitr_1.50
## [58] IRanges_2.42.0        doParallel_1.0.17      mgcv_1.9-3
## [61] rlang_1.1.6           Rcpp_1.1.0             glue_1.8.0
## [64] BiocGenerics_0.54.0   rstudioapi_0.17.1      R6_2.6.1

```